Mathematical and Computational Visualization of Hydrogenic Orbitals, Probability Density, and Quantum Flow

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Abstract

The hydrogen atom occupies a central position in quantum mechanics as one of the few nontrivial physical systems whose Schrödinger equation admits an exact analytical solution. The resulting wavefunctions generate the familiar atomic orbitals encountered throughout atomic physics and quantum chemistry.
This document introduces the mathematical foundations of hydrogenic orbitals and presents the *Atom* project as a computational framework for sampling, visualizing, and exploring these quantum states.

1 Introduction

The idea for the Atom project originated from a C++ atom simulation created by Kavan and presented on his YouTube channel [6]. The original project demonstrated how modern graphics and simulation techniques can be used to represent atomic systems in an intuitive visual way. While working with and expanding upon these ideas, a natural question emerged:

> What mathematical objects are actually being visualized when we draw an atomic orbital?

The Atom project was developed as an attempt to answer this question. Rather than treating orbitals simply as graphical structures, the project approaches them as exact solutions of the hydrogen Schrödinger equation and investigates how their mathematical properties give rise to observable geometric patterns.
The hydrogen atom provides an ideal starting point because it is one of the few physically meaningful quantum systems whose wavefunctions can be determined analytically [5,7]. These solutions introduce three quantum numbers,

$$n, \qquad l, \qquad m,$$

which completely characterize a hydrogenic orbital.

Why the quantum numbers are important. A central objective of this project is to investigate the role played by each quantum number and to understand how changes in these quantities affect the resulting orbital.
The principal quantum number

$$n$$

determines the energy level and characteristic size of the orbital. Increasing n generally produces larger orbitals and introduces additional radial structure.
The angular momentum quantum number

$$l$$

determines the overall shape of the orbital. Different values of l generate the familiar orbital families, including s-, p-, d-, and f-orbitals.
The magnetic quantum number

$$m$$

controls how the angular structure behaves around the chosen symmetry axis. Although often introduced as an orientation parameter, it also plays an important role in angular momentum, wavefunction phase, and probability flow.
By systematically varying these parameters, the project explores how the mathematical structure of quantum states is reflected in the geometry and dynamics of orbital visualizations.

Project objectives. The Atom project combines analytical quantum mechanics, numerical sampling, and real-time graphics. Its goal is not merely to display orbital shapes, but to understand how those shapes arise from the underlying mathematics.
Particular attention is given to the relationship between

- wavefunctions,
- probability density,
- orbital geometry,
- angular momentum,
- and probability current.

Studying these quantities together provides a bridge between the abstract equations of quantum mechanics and the geometric structures commonly used to represent atomic orbitals.

Software structure. The project consists of two principal computational components.

1. A Python implementation (`schrodinger.py`) that evaluates hydrogenic wavefunctions and generates samples from their probability densities.
2. A collection of OpenGL and ray-tracing visualizers written in C++ that render orbital clouds and probability-flow structures in three dimensions.

The purpose of these notes is therefore twofold:

1. To develop the mathematical foundations of hydrogenic orbitals and their associated quantum numbers.
2. To explain how these mathematical structures are translated into computational visualizations through the Atom software.

2 Historical and Physical Background

The hydrogen atom occupies a special place in physics. It is the simplest atomic system found in nature, consisting of a single electron bound to a single proton through the Coulomb force. Historically, one of the first successful models of the atom was proposed by Niels Bohr. [1] [1] In the Bohr model the electron moves in circular orbits around the nucleus, and each allowed orbit corresponds to a specific energy.
Although remarkably successful for hydrogen, the Bohr model has an important limitation: it treats the electron as a classical particle moving along a well-defined trajectory.
Modern quantum mechanics replaces this picture with a fundamentally different description. Rather than assigning the electron a precise position and path through space, quantum theory describes the electron through a wavefunction

$$\psi(\mathbf{r}, t),$$

which assigns a complex amplitude to every point in space.

From trajectories to probabilities. A natural question then arises:

Where is the electron?

In classical mechanics this question has a definite answer. If the position and velocity of a particle are known, its future motion can be determined from Newton's laws.
Quantum mechanics introduces a different perspective. The wavefunction itself is not directly observable. According to the Born interpretation, the quantity

$$|\psi(\mathbf{r}, t)|^2$$

is interpreted as a probability density.
Regions where this quantity is large correspond to locations at which an electron is more likely to be observed during a measurement, while regions where it is small correspond to locations where an observation is less likely.

Orbitals. The stationary-state solutions of the hydrogen Schrödinger equation are known as *atomic orbitals*.
Mathematically, an orbital is a solution of the Schrödinger equation for an electron moving in the Coulomb potential of the nucleus. The orbital therefore represents a quantum state rather than a trajectory.
The familiar shapes encountered in chemistry—spherical s-orbitals, dumbbell-shaped p-orbitals, cloverleaf d-orbitals, and the more complex f-orbitals—are visual representations of the probability density

$$|\psi|^2.$$

Connection to this project. The visualizations developed in the Atom project are based directly on these hydrogenic wavefunctions. Every rendered particle cloud is constructed from samples drawn from the probability density and every orbital shape ultimately originates from the mathematical solutions of the hydrogen Schrödinger equation.
To understand how these orbitals arise, it is therefore necessary to begin with the equation from which they are derived.

[1] Bohr's model represented a major step forward because it explained why atomic spectra contain discrete lines rather than a continuous range of frequencies. Although superseded by quantum mechanics, several of its ideas reappear in the modern theory of the hydrogen atom.

3 The Schrödinger Equation

The central mathematical object underlying the present project is the Schrödinger equation. The hydrogen atom is one of the few physically interesting quantum systems for which the equation admits an exact analytical solution. The orbitals visualized throughout the project are constructed directly from these solutions.

3.1 Quantum States and Wavefunctions

In classical mechanics, the state of a particle is described by its position and velocity. Quantum mechanics takes a different approach. Instead of specifying a trajectory, the state of a system is described by a wavefunction

$$\psi(\mathbf{r}, t),$$

which depends upon position and time.
Unlike a classical trajectory, a wavefunction does not specify the exact location of a particle. Instead, it contains all information that can be known about the system.

Born interpretation. The physical meaning of the wavefunction is provided by the Born rule. [2] [2, 5]
If

$$\psi(\mathbf{r}, t)$$

denotes the state of an electron, then

$$\rho(\mathbf{r}, t) = |\psi(\mathbf{r}, t)|^2 \tag{1}$$

represents the probability density of observing the electron at position $\mathbf{r}$.
The probability of finding the electron inside a region V is therefore

$$P(V) = \int_V |\psi(\mathbf{r}, t)|^2 \, dV. \tag{2}$$

Normalization. Since the electron must be found somewhere in space, probabilities must sum to one.
Consequently,

$$\int_{\mathbb{R}^3} |\psi(\mathbf{r}, t)|^2 \, dV = 1. \tag{3}$$

This requirement is known as normalization.

3.2 The Time-Dependent Schrödinger Equation

The fundamental dynamical law of non-relativistic quantum mechanics [5, 8] is

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi. \tag{4}$$

[2]The interpretation of $|\psi|^2$ as a probability density was proposed by Max Born in 1926 and remains one of the foundational principles of quantum mechanics.

Interpretation. In classical mechanics Newton's laws determine how a particle trajectory evolves through time.

In quantum mechanics the Schrödinger equation plays an analogous role.

Rather than evolving a trajectory, however, it evolves an entire probability amplitude field.

3.3 The Hydrogen Hamiltonian

The hydrogen atom [3] consists of a single electron interacting with a proton through the Coulomb force.

The corresponding Hamiltonian [5,7] is

$$\hat{H} = -\frac{\hbar^2}{2m_e}\nabla^2 - \frac{e^2}{4\pi\varepsilon_0 r}. \tag{5}$$

The Laplacian term (i.e. first term) represents kinetic energy while the Coulomb term (i.e. second term) represents electrostatic attraction between proton and electron.

Spherical symmetry. The Coulomb potential depends only on

$$r = \sqrt{x^2 + y^2 + z^2},$$

and therefore possesses spherical symmetry. In other words, the potential depends only on the distance r from the nucleus.

Because of this spherical symmetry, spherical coordinates become the natural coordinate system for solving the problem.

3.4 Stationary States

For a time-independent Hamiltonian one may seek solutions of the form

$$\psi(\mathbf{r}, t) = \psi(\mathbf{r})e^{-iEt/\hbar}. \tag{6}$$

Substitution into the time-dependent equation yields

$$\hat{H}\psi = E\psi. \tag{7}$$

This is an eigenvalue problem.

The allowed values of E correspond to the energy levels of the atom, while the corresponding eigenfunctions are the atomic orbitals.

3.5 Separation of Variables in Spherical Coordinates

Because the Coulomb potential depends only on the radial distance

$$r = \sqrt{x^2 + y^2 + z^2},$$

it is natural to express the Schrödinger equation in spherical coordinates (r, θ, ϕ).

The Laplacian takes the form

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2}{\partial \phi^2}. \tag{8}$$

Substituting this expression into the stationary Schrödinger equation

[3] The hydrogen atom is often the first quantum system studied because it is one of the few physically important systems whose Schrödinger equation admits an exact analytical solution.

$$\hat{H}\psi = E\psi$$

gives

$$-\frac{\hbar^2}{2m_e}\nabla^2\psi - \frac{e^2}{4\pi\varepsilon_0 r}\psi = E\psi. \tag{9}$$

To solve this equation, one seeks separable solutions of the form

$$\psi(r,\theta,\phi) = R(r)\Theta(\theta)\Phi(\phi). \tag{10}$$

After substitution and division by $R\Theta\Phi$, the equation separates into independent radial and angular parts.

3.6 The Azimuthal Equation

The dependence on the angle ϕ satisfies

$$\frac{d^2\Phi}{d\phi^2} = -m^2\Phi, \tag{11}$$

whose solutions are

$$\Phi(\phi) = e^{im\phi}. \tag{12}$$

Since the wavefunction must be single-valued, we require

$$\Phi(\phi + 2\pi) = \Phi(\phi).$$

Consequently,

$$e^{2\pi im} = 1,$$

which implies

$$m \in \mathbb{Z}. \tag{13}$$

The integer m is known as the magnetic quantum number.

3.7 The Polar Equation

The dependence on the polar angle θ satisfies

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \left(l(l+1) - \frac{m^2}{\sin^2\theta}\right)\Theta = 0. \tag{14}$$

This equation is known as the associated Legendre equation. Its regular solutions are the associated Legendre functions

$$P_l^m(\cos\theta), \tag{15}$$

where l is a non-negative integer satisfying

$$l \geq |m|.$$

Combining the solutions of the θ- and ϕ-equations yields the spherical harmonics

$$Y_l^m(\theta,\phi) = N_{lm}P_l^m(\cos\theta)e^{im\phi}, \tag{16}$$

where N_{lm} is a normalization constant.

The spherical harmonics form an orthonormal basis for square-integrable functions on the unit sphere and describe the angular structure of the hydrogenic wavefunctions.

3.8 The Radial Equation

After separation of variables, the radial component satisfies

$$\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\left[\frac{2m_e r^2}{\hbar^2}\left(E+\frac{e^2}{4\pi\varepsilon_0 r}\right)-l(l+1)\right]R=0. \tag{17}$$

Introducing the dimensionless coordinate

$$\rho=\frac{2r}{na_0}, \tag{18}$$

where a_0 denotes the Bohr radius, transforms the equation into a generalized Laguerre equation.

Requiring the solutions to remain finite and square-integrable leads to the normalized radial functions

$$R_{nl}(r)=N_{nl}e^{-\rho/2}\rho^l L_{n-l-1}^{2l+1}(\rho), \tag{19}$$

where

$$L_{n-l-1}^{2l+1}$$

denotes an associated Laguerre polynomial and N_{nl} is a normalization constant.

The requirement that the radial solution terminates as a polynomial introduces the principal quantum number

$$n=1,2,3,\ldots.$$

3.9 Hydrogenic Wavefunctions and Quantum Numbers

Combining the radial and angular solutions yields the complete hydrogenic eigenfunctions

$$\psi_{nlm}(r,\theta,\phi)=R_{nl}(r)Y_l^m(\theta,\phi). \tag{20}$$

The solution is characterized by three quantum numbers,

$$n,\qquad l,\qquad m,$$

satisfying

$$n=1,2,3,\ldots,$$

$$0\le l\le n-1,$$

and

$$-l\le m\le l.$$

These quantum numbers arise naturally from the requirement that the solutions of the separated differential equations remain finite, single-valued, and square-integrable.

3.10 Connection to the Implementation

The central equation of the simulation is

$$\psi_{nlm}(r, \theta, \phi) = R_{nl}(r) Y_l^m(\theta, \phi). \tag{21}$$

The radial factor

$$R_{nl}(r) = N_{nl} e^{-\rho/2} \rho^l L_{n-l-1}^{2l+1}(\rho) \tag{22}$$

is evaluated numerically using SciPy's implementation of the associated Laguerre polynomials. Similarly, the angular component

$$Y_l^m(\theta, \phi) = N_{lm} P_l^m(\cos\theta) e^{im\phi} \tag{23}$$

is evaluated using SciPy's spherical harmonic routines [9].

The complete orbital amplitude used throughout the project is therefore generated directly from the analytical solution of the hydrogen Schrödinger equation.

4 Hydrogenic Wavefunctions

Having derived the separated solutions of the hydrogen Schrödinger equation, we now examine their mathematical structure in greater detail. These solutions are known as *hydrogenic wavefunctions* and describe all bound states of an electron in the Coulomb potential of the nucleus.

The general solution obtained in the previous section is

$$\psi_{nlm}(r,\theta,\phi) = R_{nl}(r)Y_l^m(\theta,\phi), \tag{24}$$

where

- $R_{nl}(r)$ is the radial component,
- $Y_l^m(\theta,\phi)$ is the angular component,
- n, l, m are the quantum numbers arising from the separation of variables.

This decomposition is particularly useful because it separates the dependence on distance from the nucleus from the dependence on direction.

4.1 The Radial Component

The normalized radial functions are given by

$$R_{nl}(r) = N_{nl}e^{-\rho/2}\rho^l L_{n-l-1}^{2l+1}(\rho), \tag{25}$$

where

$$\rho = \frac{2r}{na_0}, \tag{26}$$

a_0 denotes the Bohr radius and L_{n-l-1}^{2l+1} is an associated Laguerre polynomial.

The radial wavefunctions form a family of square-integrable solutions of the radial Schrödinger equation and determine how the probability distribution varies with distance from the nucleus.

Several qualitative properties follow directly from the explicit form.

- The exponential factor $e^{-\rho/2}$ ensures decay as $r \to \infty$.
- The factor ρ^l determines the behaviour near the origin.
- The Laguerre polynomial introduces oscillations and radial nodes.

Since the polynomial degree is

$$n - l - 1,$$

the number of radial nodes is

$$N_{\text{radial}} = n - l - 1. \tag{27}$$

Consequently, increasing the principal quantum number generally produces increasingly rich radial structure.

Connection to the implementation. The function

```
R(n,l,r)
```

implemented in `schrodinger.py` evaluates the radial wavefunction numerically using SciPy's implementation of the associated Laguerre polynomials.

4.2 Radial Probability Distributions

A common misconception is that $R_{nl}(r)$ itself represents the probability of finding an electron at radius r.

This is not correct.

The probability density is

$$\rho(r, \theta, \phi) = |\psi_{nlm}(r, \theta, \phi)|^2. \tag{28}$$

In spherical coordinates the volume element is

$$dV = r^2 \sin\theta \, dr \, d\theta \, d\phi. \tag{29}$$

The probability of finding the electron in a spherical shell between r and $r + dr$ is therefore obtained by integrating over all angular directions,

$$P(r) \, dr = r^2 |R_{nl}(r)|^2 \, dr. \tag{30}$$

The additional factor r^2 originates entirely from the Jacobian of the spherical-coordinate transformation.

This distinction is important because maxima of the radial wavefunction need not coincide with maxima of the radial probability distribution.

For example, in the ground state ($n = 1, l = 0$), the wavefunction attains its largest value at the nucleus, whereas the radial probability distribution reaches its maximum near one Bohr radius.

4.3 The Angular Component

The angular dependence of the hydrogenic wavefunctions is described by the spherical harmonics

$$Y_l^m(\theta, \phi). \tag{31}$$

As shown in the previous section, these functions may be written as

$$Y_l^m(\theta, \phi) = N_{lm} P_l^m(\cos\theta) e^{im\phi}, \tag{32}$$

where P_l^m denotes the associated Legendre function and N_{lm} is a normalization constant.

The spherical harmonics satisfy the orthonormality relation

$$\int_0^{2\pi} \int_0^{\pi} Y_l^m(\theta, \phi)^* Y_{l'}^{m'}(\theta, \phi) \sin\theta \, d\theta \, d\phi = \delta_{ll'} \delta_{mm'}. \tag{33}$$

Consequently, the family

$$\{Y_l^m\}$$

forms an orthonormal basis of $L^2(S^2)$, the space of square-integrable functions on the unit sphere.

The role of l. The angular momentum quantum number l determines the complexity of the angular structure and fixes the number of angular nodes,

$$N_{\text{angular}} = l. \tag{34}$$

Larger values of l therefore produce increasingly complicated angular patterns.

The role of m. The magnetic quantum number m appears through the phase factor

$$e^{im\phi}.$$

Although this factor does not affect the probability density directly, it determines the phase winding of the wavefunction about the symmetry axis.

This phase structure plays an important role in angular momentum, probability current, and the probability-flow visualizations developed later in this report.

Connection to the implementation. The angular factor

$$Y_l^m(\theta, \phi) = N_{lm} P_l^m(\cos\theta) e^{im\phi} \tag{35}$$

is evaluated numerically using SciPy's implementation of spherical harmonics. In `schrodinger.py` this corresponds to

```
Y = sph_harm(m, l, phi, theta)
```

which computes the complex-valued spherical harmonic $Y_l^m(\theta, \phi)$. Combined with the radial factor $R_{nl}(r)$, this yields the complete hydrogenic wavefunction

$$\psi_{nlm}(r, \theta, \phi) = R_{nl}(r) Y_l^m(\theta, \phi).$$

4.4 Nodes

A node is a point or surface on which the wavefunction vanishes,

$$\psi_{nlm}(r, \theta, \phi) = 0. \tag{36}$$

Since the probability density is proportional to

$$|\psi_{nlm}|^2,$$

nodes correspond to regions in which the electron cannot be observed.

The nodal structure decomposes naturally into radial and angular contributions.

The number of radial nodes is

$$N_{\text{radial}} = n - l - 1, \tag{37}$$

while the number of angular nodes is

$$N_{\text{angular}} = l. \tag{38}$$

Hence the total number of nodes is

$$N_{\text{total}} = (n - l - 1) + l = n - 1. \tag{39}$$

This relation provides a direct connection between the quantum numbers and the geometric complexity of the corresponding orbital.

4.5 Connection to Orbital Visualization

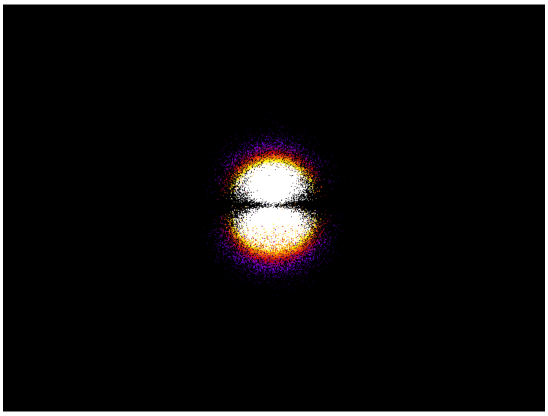
The quantity ultimately visualized by the Atom project is the probability density

$$\rho(r, \theta, \phi) = |\psi_{nlm}(r, \theta, \phi)|^2. \tag{40}$$

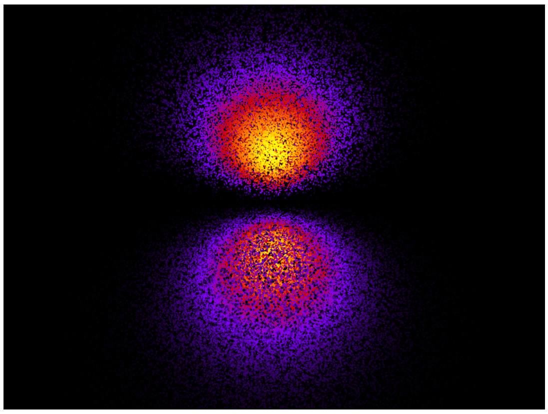
The Python generator samples points from this distribution and stores the resulting dataset for later rendering.

Consequently, the displayed orbital clouds are not arbitrary geometric objects. Rather, they constitute numerical approximations of the exact hydrogenic probability distributions obtained from the analytical solutions of the Schrödinger equation.

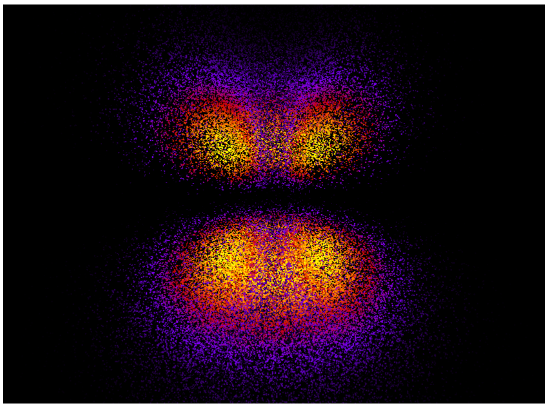
Representative examples of orbitals generated by the renderer are shown in Figure 1. As the quantum numbers increase, the probability density develops increasingly rich radial and angular structure.



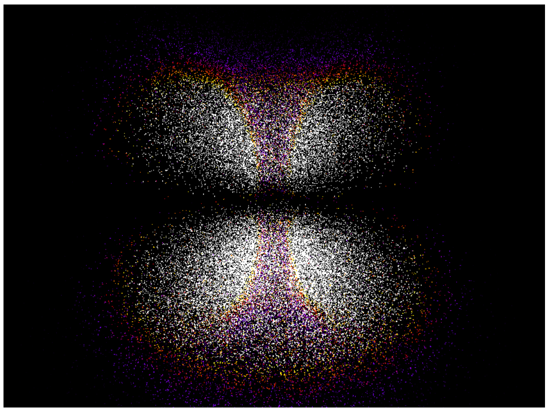
(a) $n = 1, l = 0, m = 0$



(b) $n = 2, l = 1, m = 0$



(c) $n = 3, l = 2, m = 1$



(d) $n = 4, l = 3, m = 2$

Figure 1: Hydrogenic orbital probability densities generated using the Atom visualization framework. The displayed point clouds are sampled from the probability density $|\psi_{nlm}|^2$. Increasing values of the quantum numbers produce larger orbitals, additional nodal structures, and increasingly complex angular geometry.

5 Sampling Quantum Orbitals

The solutions of the hydrogen Schrödinger equation provide an exact description of the electron probability density. While these functions can be evaluated analytically, they are defined continuously over all of three-dimensional space and therefore cannot be displayed directly.
This creates the central computational problem of the project:

> How can a continuous probability distribution be converted into a finite set of points that can later be visualized?

The purpose of `schrodinger.py` is to solve this problem. Rather than drawing orbitals from predefined geometric templates, the program generates points whose spatial distribution follows the hydrogenic probability density. The resulting point cloud therefore emerges from the mathematics itself rather than from any manually specified shape.

5.1 The Target Distribution

The starting point is the hydrogenic wavefunction

$$\psi_{nlm}(r, \theta, \phi) = R_{nl}(r) Y_l^m(\theta, \phi). \tag{41}$$

According to the Born interpretation, the corresponding probability density is

$$\rho(r, \theta, \phi) = |\psi_{nlm}(r, \theta, \phi)|^2. \tag{42}$$

Substituting the orbital decomposition gives

$$\rho(r, \theta, \phi) = |R_{nl}(r)|^2 |Y_l^m(\theta, \phi)|^2. \tag{43}$$

This density defines the distribution that the sampling algorithm seeks to reproduce.

Interpretation. If a region of space possesses a large value of

$$\rho,$$

many sampled points should appear in that region. Likewise, regions of small probability density should contain relatively few points.
As the number of samples increases, the resulting cloud should converge toward the underlying quantum probability distribution.

5.2 The Sampling Problem

Hydrogenic probability densities are mathematically complicated objects. They contain exponential terms, Laguerre polynomials, spherical harmonics, and multiple nodal surfaces.
Directly sampling from such a distribution is difficult. A numerical method is therefore required.

Desired property. Suppose that a region V contains probability

$$P(V).$$

If

$$N$$

points are generated, then approximately

$$NP(V)$$

points should lie inside that region.
The objective of the sampling algorithm is to produce a finite set of points satisfying this property as accurately as possible.

5.3 Monte Carlo Rejection Sampling

To generate orbital samples, the project uses a Monte Carlo method known as rejection sampling. [4]

The basic idea is simple. Candidate points are generated from an easily sampled distribution and are then accepted or rejected according to the value of the quantum probability density at that location.

For a candidate point

$$(r, \theta, \phi),$$

the algorithm computes

$$\rho(r, \theta, \phi) = |\psi_{nlm}(r, \theta, \phi)|^2.$$

A uniformly distributed random number

$$u \in [0, 1]$$

is then generated and the candidate is accepted whenever

$$u < \frac{\rho}{\rho_{\max}}. \tag{44}$$

Why this works. Locations with high probability density have a larger chance of being accepted, while locations with low probability density are more likely to be rejected.
Over many iterations, the accepted samples reproduce the desired quantum distribution.

5.4 Sampling Strategy

The implementation does not propose candidate points uniformly throughout space.

Instead, it uses proposal distributions that roughly match the expected geometry of hydrogen orbitals.

Radial coordinate. The radial coordinate is generated using an exponential distribution,

$$r \sim \text{Exp}(n^2). \tag{45}$$

In the implementation this appears as

```
r = np.random.exponential(scale=n**2)
```

[4] Rejection sampling is one of the simplest and most widely used Monte Carlo techniques. Although not always the most computationally efficient, it is conceptually transparent and well suited to visualization projects.

Hydrogenic wavefunctions decay approximately exponentially with distance from the nucleus, making this a natural and computationally efficient choice.
The factor

$$n^2$$

reflects the fact that the characteristic size of a hydrogen orbital grows approximately as

$$n^2 a_0.$$

Angular coordinates. The angular coordinates are proposed uniformly:

$$\theta \sim U(0, \pi), \tag{46}$$
$$\phi \sim U(0, 2\pi). \tag{47}$$

The orbital geometry is therefore not imposed during sampling. Instead, it emerges naturally through the acceptance step via the spherical harmonic contribution

$$|Y_l^m(\theta, \phi)|^2.$$

5.5 Wavefunction Evaluation

For each candidate point the program evaluates the full hydrogenic wavefunction:

$$\psi_{nlm} = R_{nl}(r) Y_l^m(\theta, \phi). \tag{48}$$

In the implementation this corresponds to

```
Y = sph_harm(m,l,phi,theta)
return R(n,l,r) * Y
```

The resulting quantity is generally complex,

$$\psi = \psi_R + i\psi_I. \tag{49}$$

While probability density depends only on

$$|\psi|^2,$$

the full complex wavefunction contains additional information related to phase, angular momentum, and probability current. For this reason both the real and imaginary components are retained.

5.6 Coordinate Conversion

Accepted samples are converted from spherical coordinates into Cartesian coordinates:

$$x = r \sin\theta \cos\phi, \tag{50}$$
$$y = r \sin\theta \sin\phi, \tag{51}$$
$$z = r \cos\theta. \tag{52}$$

These coordinates are stored together with the wavefunction data and form the basis of all subsequent visualizations.

5.7 Output Data

Each accepted sample is written to a JSON file containing

- the quantum numbers n, l, and m,
- Cartesian coordinates,
- the real component of the wavefunction,
- the imaginary component of the wavefunction.

This design separates orbital generation from rendering. The expensive quantum-mechanical calculations are performed once during sampling, while the resulting dataset can be visualized repeatedly without recomputing the wavefunction.

5.8 Statistical Interpretation

The generated point cloud should be understood as a numerical approximation of the probability density

$$|\psi_{nlm}|^2.$$

As the number of samples increases,

$$N \to \infty,$$

the empirical distribution of points converges toward the underlying quantum probability distribution.

Practical implications. Small sample counts produce noisy orbital clouds, while large sample counts reveal finer geometric detail, sharper nodal structures, and a more accurate representation of the wavefunction.

Summary. The role of `schrodinger.py` is to transform an analytical solution of the hydrogen Schrödinger equation into a finite set of samples that can be processed computationally. Through rejection sampling, the generated point cloud reproduces the probability density of the selected orbital and provides the numerical foundation for the visualization stage of the Atom project. The next step is to convert these sampled points into an observable three-dimensional structure. This is the task of the rendering system.

6 Rendering Hydrogenic Orbitals

The previous section described how the continuous probability density of a hydrogenic wavefunction is converted into a finite set of sampled points. At this stage, however, the output consists only of numerical data. Although the point cloud contains the information required to describe the orbital, its structure is not immediately apparent.
The purpose of the rendering system is therefore to transform these samples into a visual representation that can be explored and analysed.

Separation of tasks. The project deliberately separates orbital generation from visualization.

1. The Python implementation evaluates the wavefunction and generates samples from the probability density.
2. The C++ implementation is responsible for displaying and exploring the resulting orbital.

This separation ensures that the mathematics of the hydrogen atom remains independent of the graphics implementation.

6.1 Point Clouds as Probability Distributions

Suppose that

$$\{\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N\}$$

denotes the collection of sampled points generated by the algorithm described in the previous section.

If these points are drawn from the distribution

$$\rho(\mathbf{r}) = |\psi(\mathbf{r})|^2,$$

then regions containing many points correspond to regions of high probability density, while sparse regions correspond to regions of low probability density.
As the number of samples increases, the empirical distribution of the point cloud approaches the underlying quantum probability density.

Interpretation. The rendered particles do not represent individual electrons. [5]

Instead, they should be interpreted as samples from many hypothetical measurements of the same quantum state.

The orbital cloud is therefore a visual approximation of the probability distribution rather than a physical trajectory.

6.2 Visualizing Orbital Geometry

One of the primary goals of the renderer is to make the geometry of the wavefunction visible.

Hydrogenic orbitals contain structures that are difficult to appreciate from equations alone, including

- nodal planes,
- angular lobes,
- radial shells,

[5]A common misconception is that orbital visualizations depict electrons moving through space. In reality they represent the statistical outcome of many hypothetical measurements of the same quantum state.

- symmetry axes.

When displayed as a three-dimensional point cloud, these structures become immediately recognizable.

Nodes. A node is a region where

$$\psi = 0.$$

Since the probability density satisfies

$$|\psi|^2 = 0,$$

such regions contain no sampled particles.

Consequently nodal structures appear naturally as empty regions within the rendered cloud.

6.3 Interactive Exploration

Hydrogenic orbitals are fundamentally three-dimensional objects.

For this reason a static two-dimensional image often obscures important geometric information.

The visualization therefore employs an orbital camera system in which the observer moves around the orbital rather than moving the orbital itself.
This allows the same quantum state to be viewed from multiple perspectives and makes features such as nodal surfaces and radial structure much easier to identify.

Camera coordinates. The camera position is described in spherical coordinates by

$$\mathbf{r}_{\text{camera}} = \begin{pmatrix} R\sin\theta_c \cos\phi_c \\ R\cos\theta_c \\ R\sin\theta_c \sin\phi_c \end{pmatrix}. \tag{53}$$

This representation is particularly convenient because it mirrors the spherical symmetry of the hydrogen atom itself.

6.4 Color and Probability Density

Geometry alone does not always reveal the full structure of an orbital.

The renderer therefore assigns colors according to quantities derived from the wavefunction. In practice, a scalar quantity

$$I(r, \theta, \phi)$$

is computed from the orbital and mapped to a color function

$$C = f(I). \tag{54}$$

This provides additional visual information regarding the distribution of probability throughout the orbital.

Purpose of color. Color is used to emphasize

- regions of high probability density,
- regions of low probability density,
- radial structure,
- transitions near nodal surfaces.

Several implementations of the project use inferno-style color maps, which provide strong contrast across a wide range of values.

6.5 Ray-Traced Orbital Rendering

In addition to conventional OpenGL rendering, the project includes a ray-traced visualization mode.

Rather than projecting geometry directly onto the screen, the ray tracer simulates rays emitted from the camera and determines their interaction with the orbital particles.

Advantages. Ray tracing provides several benefits for scientific visualization:

- improved depth perception,
- realistic shadows,
- enhanced spatial understanding,
- clearer separation of overlapping structures.

Although the underlying orbital remains unchanged, improved lighting can make subtle geometric features substantially easier to recognize.

6.6 Beyond Probability Density

Most textbook orbital diagrams display only the quantity

$$|\psi|^2.$$

This is sufficient for understanding orbital shape, but it does not capture the full information contained within the wavefunction.
The complete wavefunction

$$\psi = \psi_R + i\psi_I$$

also contains phase information. While phase is not immediately visible in ordinary orbital diagrams, it becomes important when discussing angular momentum and probability current. For this reason the project preserves the complete complex wavefunction during the sampling stage, allowing later sections to investigate probability flow in addition to probability density.

Summary. The rendering system transforms sampled orbital data into an observable three-dimensional structure. Point clouds provide a geometric approximation of the quantum probability density, while camera controls, color mapping, and ray tracing reveal features that would otherwise be difficult to extract from the analytical wavefunction alone. The renderer therefore serves as a bridge between the mathematical solution of the Schrödinger equation and its geometric interpretation.
Having established how orbital densities are visualized, we now turn to a more subtle question: how quantum probability itself can exhibit circulation and flow.

7 Probability Current and Quantum Flow

Up to this point, the discussion has focused primarily on probability density.
The quantity

$$\rho(\mathbf{r}) = |\psi(\mathbf{r})|^2$$

describes where an electron is likely to be observed if a measurement is performed.
However, probability density does not fully describe a quantum state.
Two wavefunctions may possess identical probability distributions while exhibiting fundamentally different dynamical behavior.

A natural question. If probability is distributed throughout space, can that probability also flow?
Quantum mechanics answers this question in the affirmative.
The mathematical object describing such flow is known as the *probability current density.*

7.1 Motivation from Classical Physics

Consider a fluid occupying a region of space.
The density of the fluid may be described by a function

$$\rho(\mathbf{r}, t).$$

If the fluid moves with velocity field

$$\mathbf{v}(\mathbf{r}, t),$$

then the associated flow is described by the current [5,7]

$$\mathbf{J} = \rho\mathbf{v}. \tag{55}$$

Conservation of mass. Matter is not created or destroyed during ordinary fluid motion.
Consequently the density must satisfy

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0. \tag{56}$$

This is known as the continuity equation.

Quantum analogy. Probability behaves in an analogous manner.
The total probability of finding an electron must remain equal to one.
If probability leaves one region of space, it must appear elsewhere.
Quantum mechanics therefore possesses its own continuity equation and its own notion of current.

7.2 Derivation of Probability Current

The time-dependent Schrödinger equation is

$$i\hbar\frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi + V\psi. \tag{57}$$

Taking the complex conjugate yields

$$-i\hbar\frac{\partial \psi^*}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi^* + V\psi^*. \tag{58}$$

A straightforward calculation shows that

$$\frac{\partial|\psi|^2}{\partial t} + \nabla \cdot \mathbf{j} = 0, \tag{59}$$

where

$$\mathbf{j} = \frac{\hbar}{2mi}\left(\psi^*\nabla\psi - \psi\nabla\psi^*\right). \tag{60}$$

The vector

$$\mathbf{j}$$

is called the probability current density.

Interpretation. Just as a river possesses a direction and magnitude of flow, probability current describes the direction and magnitude of quantum probability transport through space.

7.3 Density Versus Current

A useful analogy is provided by weather maps.

Density. A weather map may display atmospheric pressure.
This indicates where air is concentrated.

Flow. A wind map displays how air moves.
This indicates the direction of transport.

Quantum interpretation. Similarly,

$$|\psi|^2$$

describes where probability is concentrated,
while

$$\mathbf{j}$$

describes how probability moves.
Both quantities are important.
A complete description of a quantum state requires knowledge of both.

7.4 Phase and Probability Flow

The probability current depends not only on the magnitude of the wavefunction but also on its phase.
Suppose a wavefunction is written as

$$\psi = Ae^{iS}, \tag{61}$$

where

- A is the amplitude,
- S is the phase.

Substitution into the current equation yields

$$\mathbf{j} = \frac{\hbar}{m} A^2 \nabla S. \tag{62}$$

Since

$$A^2 = |\psi|^2,$$

the current becomes

$$\mathbf{j} = \frac{\hbar}{m} |\psi|^2 \nabla S. \tag{63}$$

Important consequence. Probability flow is determined by gradients in phase.

If the phase does not vary spatially, the probability current vanishes.

If the phase winds through space, a nonzero current appears.

This observation is the key to understanding the vortex structures that appear later in the project.

7.5 Probability Current in Hydrogen Orbitals

Recall that hydrogenic orbitals are written as

$$\psi_{nlm} = R_{nl}(r) Y_l^m(\theta, \phi). \tag{64}$$

The spherical harmonics contain the factor

$$e^{im\phi}. \tag{65}$$

This term introduces an angle-dependent phase.

Case $m = 0$. For

$$m = 0,$$

the phase factor becomes

$$e^{i0\phi} = 1.$$

No azimuthal phase winding occurs.

Consequently the orbital possesses no preferred direction of circulation about the symmetry axis.

Case $m \neq 0$. For

$$m \neq 0,$$

the phase changes continuously as the angle ϕ changes.

The wavefunction therefore acquires an azimuthal structure that can support circulating probability current.

Geometric picture. One may imagine the phase wrapping around the orbital axis.

As one completes a full revolution around the nucleus, the phase undergoes a corresponding rotation.

This winding generates a natural tendency for probability flow to follow an azimuthal direction.

7.6 Physical Intuition: From Static Orbitals to Quantum Circulation

The distinction between the states

$$m = 0$$

and

$$m = \pm 1$$

is often difficult to appreciate from equations alone.

A useful way to think about these states is to distinguish between a wavefunction that possesses no azimuthal circulation and one that contains a nontrivial phase winding around the orbital axis.

The case $m = 0$. Consider first a hydrogen orbital whose magnetic quantum number is zero.

In this situation the angular factor

$$e^{im\phi}$$

reduces to

$$e^{i0\phi} = 1.$$

The wavefunction therefore possesses no angular phase winding around the symmetry axis. The probability density may still exhibit a rich geometric structure, but there is no preferred sense of circulation around the axis.
Within the visualization developed in this project, the corresponding probability-flow field vanishes and the orbital appears stationary.

The case $m = \pm 1$. For nonzero magnetic quantum number,

$$m = \pm 1,$$

the situation changes significantly.

The phase factor becomes

$$e^{\pm i\phi},$$

which continuously changes as one moves around the orbital axis.
A complete revolution around the axis corresponds to a complete rotation of the phase.

This introduces a natural sense of circulation into the quantum state.

An analogy with standing and rotating waves. A useful analogy comes from classical wave mechanics.

A standing wave on a string oscillates in place without transporting material along the string.

By contrast, a rotating wave pattern carries a sense of motion around its supporting geometry.
The states $m = \pm 1$ may be viewed as quantum analogues of rotating wave patterns, whereas the state $m = 0$ resembles a standing-wave configuration.

Interaction with light. In real atomic systems, transitions between states of different magnetic quantum number can occur through interactions with electromagnetic radiation.

Photons carry angular momentum in addition to energy.

Consequently, when an atom absorbs appropriately polarized light, the magnetic quantum number may change according to quantum-mechanical selection rules.[6]

From the perspective of the present project, such transitions are particularly interesting because they change both the geometry of the orbital density and the structure of the associated probability flow.

Connection to chemistry. An important distinction exists between the complex states

$$m = -1,\ 0,\ +1$$

commonly used in physics and the orbital pictures often encountered in chemistry.

The familiar orbitals [7]

$$p_x, \qquad p_y, \qquad p_z$$

are not usually eigenstates of the magnetic quantum number [3, 5].

Instead, they are linear combinations of the complex spherical harmonics.

For example, the directional orbitals used in chemistry emerge from superpositions of the rotating states

$$m = +1$$

and

$$m = -1.$$

The resulting combination eliminates the net azimuthal circulation and produces orbitals pointing in fixed spatial directions.

Why this matters. The visualization developed in this project emphasizes the underlying eigenstates labelled by m.

Consequently, features related to phase winding and orbital circulation remain visible.

By contrast, many chemistry textbooks emphasize real-valued orbital combinations, which highlight bonding geometry but conceal the circulating phase structure present in the underlying quantum states.

Conceptual summary. The transition

$$m = 0 \longrightarrow m = \pm 1$$

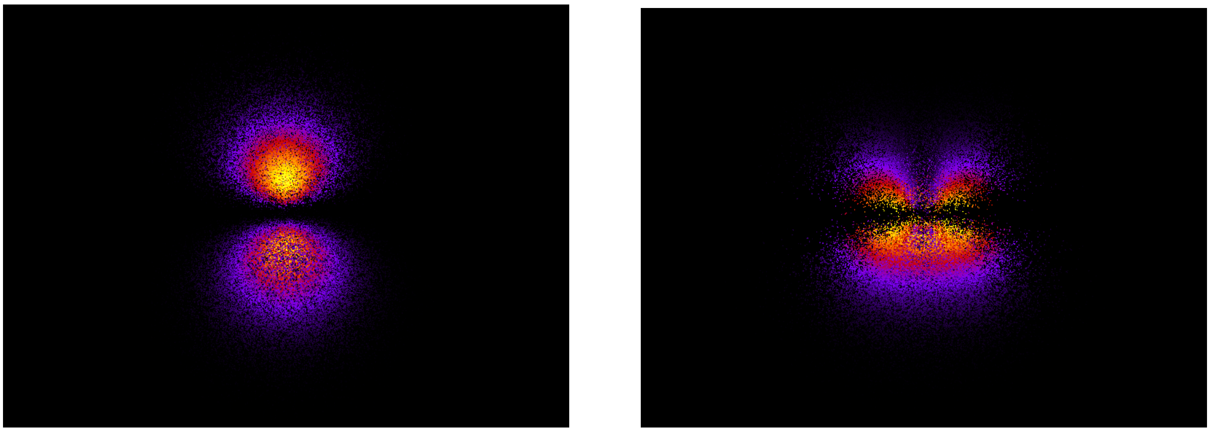
should not be interpreted simply as a change of orbital shape.

Rather, it represents the appearance of angular phase structure, orbital angular momentum, and circulating probability flow.

The observation of the vortex therefore reveales a property of the quantum state that remains hidden in most conventional orbital diagrams.

[6] A discussion of angular momentum eigenstates and selection rules is given in standard quantum-mechanics texts and in the ETH lecture notes on angular momentum [4].

[7] The orbitals commonly drawn in chemistry textbooks are usually real linear combinations of complex spherical harmonics. This makes bonding geometry easier to visualize but conceals part of the underlying phase structure.



(a) $(n, l, m) = (2, 1, 0)$ (b) $(n, l, m) = (2, 1, 1)$

Figure 2: Comparison of two hydrogenic orbitals with identical principal and angular momentum quantum numbers but different magnetic quantum numbers. Both visualizations use the same camera position, color map, particle count, and rendering parameters. The state with $m = 0$ possesses no azimuthal phase winding, whereas the state with $m = 1$ introduces angular phase structure associated with orbital angular momentum and circulating probability flow.

7.7 Angular Momentum and Flow

The magnetic quantum number is directly related to angular momentum.

Hydrogen orbitals satisfy [4, 7]

$$L_z Y_l^m = m\hbar Y_l^m. \tag{66}$$

The quantity

$$m\hbar$$

represents the angular momentum projected onto the chosen z-axis.

Physical interpretation. Nonzero values of m correspond to states possessing nonzero orbital angular momentum about the axis.

Because angular momentum involves rotational behavior, it is natural that such states exhibit circulating probability currents.

Connecting shape and flow. The orbital density determines where probability resides.

The angular momentum determines how that probability tends to circulate.

Consequently two orbitals may exhibit similar densities while possessing very different current structures.

7.8 Connection to the Atom Project

The probability current formula motivated the implementation of orbital flow visualizations within the C++ renderer.

For each particle the code computes an azimuthal velocity field associated with the magnetic quantum number.

In simplified form the implemented velocity is

$$v_\phi = \frac{\hbar m}{m_e r \sin\theta}. \tag{67}$$

Although this expression does not constitute a full numerical evaluation of the exact probability current density, it captures an important qualitative feature of hydrogenic states:

> States possessing nonzero magnetic quantum number naturally develop azimuthal circulation around the orbital axis.

As shown in Figure 3, the velocity field of a vortex follows an inverse radial dependence characteristic of quantized circulation.

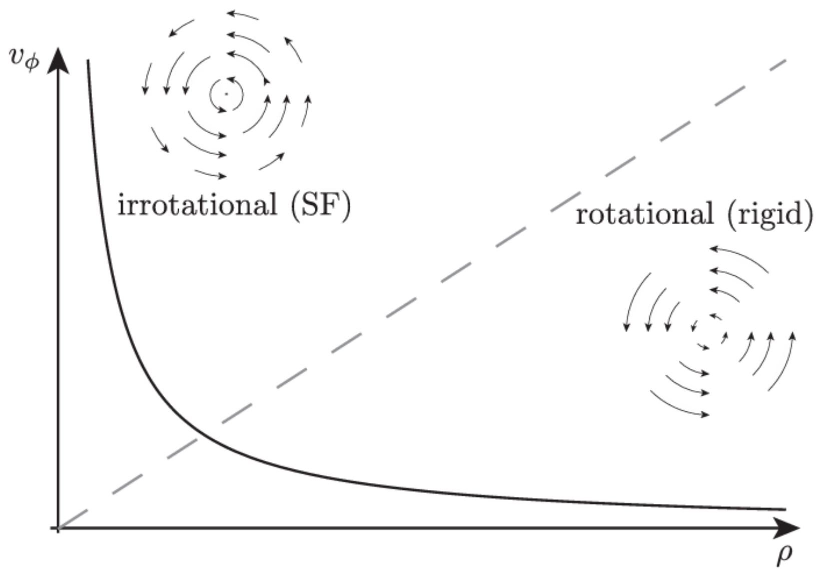


Figure 3: Azimuthal velocity profile of a quantized vortex. Unlike a rigid-body rotation, where the tangential velocity increases linearly with radius, the vortex velocity scales as $v_\phi \propto 1/\rho$ and diverges near the vortex core. This behaviour motivates the circulation model used in the probability-flow visualization.

Consequence for visualization. When

$$m = 0,$$

the velocity field vanishes.
The probability cloud remains stationary.
When

$$m = 1,$$

a nonzero circulation appears.
The resulting orbital exhibits a vortex-like structure that can be observed through camera motion.
Figure 4 illustrates the characteristic depletion of density at the center of a vortex.

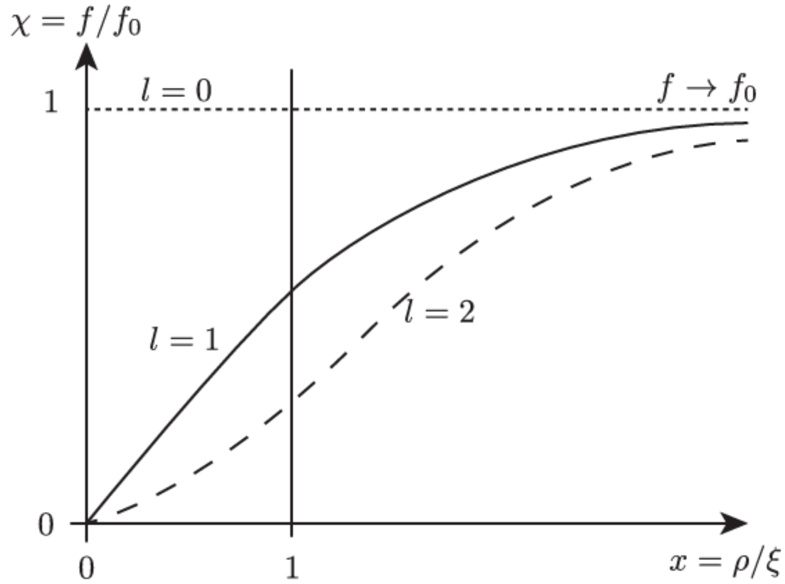


Figure 4: Density profiles for condensates containing vortices with different winding numbers. The density vanishes at the vortex core and gradually approaches its bulk value away from the rotation axis. The crossover occurs on the scale of the healing length.

7.9 Why Probability Current Matters

Many introductory orbital diagrams focus exclusively on

$$|\psi|^2.$$

While this is sufficient for understanding orbital shape, it conceals a large amount of information contained within the wavefunction.

Probability current reveals:

- phase structure,
- angular momentum,
- directional flow,
- vortex behavior,
- topological winding.

These concepts become particularly important when studying states with nonzero magnetic quantum number.

Summary. Probability density describes where quantum probability is located, whereas probability current describes how that probability moves through space. The current depends critically on wavefunction phase and therefore connects naturally to the magnetic quantum number m. This relationship ultimately explains the circulation patterns observed in the Atom visualization and provides the theoretical foundation for the vortex structures discovered during development.

8 Exploring Quantum States Through Interactive Controls

The previous section introduced the concept of probability current and showed that a quantum state is characterized not only by its probability density

$$|\psi|^2,$$

but also by its phase structure and the associated flow of probability through space.
This raises a natural question:

> How do the quantum numbers n, l, and m influence the geometry and flow properties of a hydrogenic orbital?

Rather than studying this question only through equations, the Atom project allows these quantities to be modified interactively. By varying the quantum numbers in real time and immediately observing the resulting changes, it becomes possible to develop intuition for the role played by each parameter.

A numerical laboratory for quantum mechanics. Many of the mathematical objects introduced in the previous sections are difficult to visualize directly. The wavefunction itself exists as a complex-valued function, while quantities such as angular momentum, phase winding, and probability current are often introduced abstractly.
The interactive controls provide a practical way to explore these ideas. Changing one quantum number while keeping the others fixed allows its individual contribution to the orbital to be isolated and studied.
In this sense, the renderer functions as a small numerical laboratory for the hydrogen atom. The user can directly investigate how changes in

$$n, \qquad l, \qquad m$$

affect orbital size, shape, angular momentum, and probability flow.

From mathematics to visualization. The hydrogenic wavefunction

$$\psi_{nlm} = R_{nl}(r)Y_l^m(\theta, \phi) \tag{68}$$

depends explicitly on all three quantum numbers.
Consequently, changing any one of them alters the quantum state being visualized.
The controls implemented in the Atom project therefore provide a direct way of exploring how the mathematical structure of the wavefunction is translated into observable geometric and dynamical behaviour.

8.1 Interactive Quantum Number Controls

The visualization provides direct keyboard controls for modifying the quantum numbers:

Key	Effect
W	Increase n
S	Decrease n
E	Increase l
D	Decrease l
R	Increase m
F	Decrease m

Whenever one of these values changes, the wavefunction is recomputed and the orbital cloud is regenerated.

8.2 The Principal Quantum Number n

The principal quantum number controls the energy level and characteristic size of the orbital. Increasing

$$n$$

moves the electron into higher-energy states. These states generally extend farther from the nucleus and often contain additional radial structure.

Visual intuition. A useful way to think about n is as a measure of scale.
Small values of n produce compact orbitals concentrated near the nucleus, while larger values produce increasingly extended probability clouds.

Mathematical role. The value of n appears directly in the radial wavefunction

$$R_{nl}(r),$$

and influences both the orbital radius and the number of radial nodes.
The total number of nodes satisfies

$$N_{\text{total}} = n - 1.$$

Consequently increasing n generally increases the complexity of the orbital.

Implementation. Within the code, the keys

```
W
```

and

```
S
```

modify the value of n. The updated value is then used when evaluating the radial part of the hydrogenic wavefunction.

8.3 The Angular Momentum Quantum Number l

The angular momentum quantum number controls orbital shape.
Where n determines size, l determines geometry.

Visual intuition. Changing l transforms the overall structure of the orbital.
For example,

$$l = 0$$

produces spherical s-orbitals,

$$l = 1$$

produces two-lobed p-orbitals,
and higher values generate increasingly complex structures.

Mathematical role. The value of l determines the spherical harmonic

$$Y_l^m(\theta, \phi),$$

and therefore controls the angular dependence of the wavefunction.
It also determines the number of angular nodes:

$$N_{\text{angular}} = l.$$

Implementation. The keys

```
E
```

and

```
D
```

increase and decrease the value of l, allowing the user to move between different orbital families in real time.

8.4 The Magnetic Quantum Number m

The most subtle and perhaps most interesting quantum number is

$$m.$$

Unlike n and l, which primarily affect size and shape, m is closely related to angular momentum, phase structure, and probability flow.

Visual intuition. Changing m often produces less dramatic geometric changes than altering n or l. However, it changes how the wavefunction behaves around the orbital axis.
In particular, nonzero values of m introduce an azimuthal phase factor

$$e^{im\phi}.$$

As discussed in the previous section, this phase winding is closely related to probability current and orbital circulation.

The case $m = 0$. When

$$m = 0,$$

the phase factor becomes

$$e^{i0\phi} = 1.$$

No azimuthal phase winding occurs.

The case $m \neq 0$. When

$$m = \pm 1, \pm 2, \ldots,$$

the phase varies continuously around the orbital axis.
This introduces angular momentum and allows circulating probability currents to appear.

Implementation. The keys

`R`

and

`F`

control the value of m.
The visualization uses this quantity when constructing the probability flow field

$$v_\phi = \frac{\hbar m}{m_e r \sin\theta}. \tag{69}$$

Consequently, changing m directly alters the strength and direction of the orbital circulation.

8.5 Why These Controls Matter

Taken together, the controls reveal the distinct roles of the three quantum numbers:

- n controls the scale of the orbital,
- l controls its shape,
- m controls its angular phase structure and probability flow.

Because the hydrogenic wavefunction depends on all three quantities,

$$\psi_{nlm} = R_{nl}(r) Y_l^m(\theta, \phi), \tag{70}$$

modifying any one of them changes the resulting quantum state.

Summary. The interactive controls transform the renderer from a static visualization into a tool for exploring the mathematics of the hydrogen atom. By adjusting n, l, and m directly, the user can observe how energy level, orbital geometry, angular momentum, and probability flow emerge from the structure of the wavefunction itself.

9 Software Architecture and Mathematical Implementation

The Atom project combines analytical quantum mechanics, numerical sampling, and interactive visualization into a single computational pipeline. Although the final output is a three-dimensional orbital rendering, the process begins with the exact analytical solution of the hydrogen Schrödinger equation and proceeds through several intermediate stages before visualization.

$$\text{Schrödinger Equation} \rightarrow \text{Wavefunction} \rightarrow \text{Probability Density} \rightarrow \text{Sampling} \rightarrow \text{Visualization.} \tag{71}$$

Each stage transforms the mathematical description of the quantum state into a representation that is increasingly suitable for numerical processing and graphical rendering.

9.1 Stage 1: Schrödinger Equation

The computational workflow begins with the stationary Schrödinger equation for the hydrogen atom,

$$\hat{H}\psi = E\psi. \tag{72}$$

Because the Coulomb potential is spherically symmetric, the equation can be separated into radial and angular components. This analytical treatment provides the mathematical foundation of the entire project and introduces the quantum numbers

$$n, \qquad l, \qquad m.$$

These quantum numbers uniquely determine the hydrogenic state that will later be visualized.

9.2 Stage 2: Wavefunction Evaluation

The next stage is implemented in `schrodinger.py`. For a selected set of quantum numbers, the program evaluates the hydrogenic wavefunction

$$\psi_{nlm}(r, \theta, \phi) = R_{nl}(r)\, Y_l^m(\theta, \phi), \tag{73}$$

where R_{nl} is the normalized radial function and Y_l^m is the spherical harmonic.

This step produces the complete complex-valued quantum state throughout space and forms the basis for all subsequent calculations.

Preservation of phase information. Rather than storing only magnitudes, the implementation retains both the real and imaginary parts of the wavefunction,

$$\psi = \psi_R + i\psi_I.$$

This information becomes important when visualizing angular momentum, phase winding, and probability-flow effects.

9.3 Stage 3: Probability Density Construction

The wavefunction is transformed into a measurable quantity through the Born rule. The probability density is computed as

$$\rho(r, \theta, \phi) = |\psi_{nlm}(r, \theta, \phi)|^2. \tag{74}$$

This density describes the probability distribution associated with the orbital and defines the target distribution that the sampling algorithm must reproduce.

Regions where ρ is large correspond to locations in which the electron is more likely to be observed, while regions where $\rho = 0$ correspond to nodal structures.

9.4 Stage 4: Monte Carlo Sampling

Because the probability density is a continuous function of space, it cannot be rendered directly. The density is therefore converted into a finite collection of samples using Monte Carlo rejection sampling.

Candidate points are generated in spherical coordinates and accepted according to

$$u < \frac{\rho}{\rho_{\max}}, \tag{75}$$

where u is a uniformly distributed random number.

Accepted samples are distributed according to the quantum probability density and therefore form a numerical approximation of the orbital.

Coordinate conversion. After acceptance, each sample is transformed into Cartesian coordinates,

$$x = r \sin\theta \cos\phi, \tag{76}$$

$$y = r \sin\theta \sin\phi, \tag{77}$$

$$z = r \cos\theta, \tag{78}$$

which are more convenient for rendering.

9.5 Stage 5: Data Export and Storage

The generated samples are written to a JSON data file containing

- the quantum numbers n, l, and m,
- Cartesian particle positions,
- the real part of the wavefunction,
- the imaginary part of the wavefunction.

This design separates the quantum-mechanical computation from the visualization stage. Orbital generation must only be performed once, while the resulting dataset can be rendered repeatedly without recomputing the wavefunction.

9.6 Stage 6: Visualization and Rendering

The sampled dataset is subsequently loaded by the C++ visualization system.

At this stage no additional quantum-mechanical calculations are required. Each sample is rendered as an individual particle in three-dimensional space, producing a point-cloud approximation of the underlying probability density.

Interactive exploration. The renderer allows the orbital to be viewed from arbitrary directions and under different rendering modes, making it possible to investigate features such as

- nodal planes,
- nodal cones,
- radial shells,
- symmetry axes,
- angular structure.

Many of these features are difficult to recognize directly from the analytical wavefunction.

9.7 Stage 7: Ray-Traced Rendering

In addition to real-time visualization, the project includes a ray-tracing mode.

The ray tracer simulates light transport and visibility by tracing rays through the particle cloud. Although the underlying orbital remains unchanged, realistic lighting, shadows, and depth cues often make subtle structures easier to identify.

9.8 Stage 8: Probability-Flow Visualization

A distinctive feature of the Atom project is the visualization of orbital circulation inspired by the theory of probability current.

Instead of displaying only the density $|\psi|^2$, particles may be associated with the velocity field

$$v_\phi = \frac{\hbar m}{m_e r \sin\theta}. \tag{79}$$

This model is motivated by the connection between wavefunction phase, angular momentum, and the magnetic quantum number m.

Interpretation. The displayed trajectories should not be interpreted as literal electron paths. Rather, they provide an intuitive representation of how nonzero values of m are related to phase winding and circulating probability flow.

9.9 Summary

The Python implementation evaluates hydrogenic wavefunctions and generates samples from their probability distributions, while the C++ visualization system transforms those samples into interactive orbital representations. Together, these components provide a computational framework for exploring the geometry, phase structure, and probability flow of hydrogenic quantum states.

10 Discussion, Limitations, and Future Directions

The Atom project began as an attempt to visualize hydrogenic orbitals in three dimensions. During development, however, the project gradually became an investigation of how the quantum numbers n, l, and m influence the structure of a quantum state.
By combining analytical solutions of the hydrogen Schrödinger equation with numerical sampling and interactive visualization, it became possible to explore the relationship between orbital geometry, probability density, angular momentum, and probability flow.

10.1 What the Project Represents Exactly

Several components of the project correspond directly to established quantum-mechanical theory.

Hydrogenic wavefunctions. The orbitals generated by the Python implementation are exact solutions of the hydrogen Schrödinger equation,

$$\psi_{nlm} = R_{nl}(r)Y_l^m(\theta, \phi). \tag{80}$$

The radial functions, spherical harmonics, and quantum numbers are not approximations. They arise directly from the analytical solution of the hydrogen atom.

Probability density. The probability density

$$\rho = |\psi|^2 \tag{81}$$

is likewise exact.
The point clouds generated by the Monte Carlo sampling procedure are numerical approximations of this probability distribution.

Orbital geometry. Features such as orbital lobes, nodal surfaces, radial shells, and angular structure emerge naturally from the underlying wavefunction. They are therefore consequences of the mathematics rather than artificial features introduced by the renderer.

10.2 What the Project Approximates

Not all aspects of the visualization correspond to exact physical calculations.

Probability-flow visualization. The renderer introduces the velocity field

$$v_\phi = \frac{\hbar m}{m_e r \sin\theta}, \tag{82}$$

which is motivated by the theory of probability current.
However, the implementation does not evaluate the full expression

$$\mathbf{j} = \frac{\hbar}{2mi}\left(\psi^*\nabla\psi - \psi\nabla\psi^*\right) \tag{83}$$

directly.

Interpretation. The resulting circulation should therefore be interpreted as a visualization inspired by probability current rather than an exact numerical calculation of quantum flow.
Its purpose is to provide intuition for the relationship between phase, angular momentum, and circulation.

10.3 What Was Learned

One of the most valuable outcomes of the project was the realization that orbital geometry represents only part of the information contained within a quantum state.

The role of n. The principal quantum number determines the overall energy and size of the orbital. Increasing n produces larger orbitals and more complex radial structure.

The role of l. The angular momentum quantum number determines orbital shape. Changing l transforms the geometry of the wavefunction and introduces additional angular nodes.

The role of m. The magnetic quantum number proved particularly interesting.

Although it often produces relatively subtle changes in orbital geometry, it strongly influences phase structure, angular momentum, and probability flow through the factor

$$e^{im\phi}.$$

Exploring different values of m highlighted the connection between spherical harmonics, phase winding, and circulation around the orbital axis.

Density versus phase. Perhaps the most important conceptual insight gained during the project was the distinction between probability density and phase.

A visualization based solely on

$$|\psi|^2$$

shows where the electron is likely to be found.

The complete wavefunction

$$\psi = \psi_R + i\psi_I$$

contains additional information that cannot be recovered from the density alone.
This additional information becomes visible through probability current and the associated flow structures introduced in the visualization.

10.4 Computational Limitations

Several limitations remain.

Finite sampling. All orbital clouds are constructed from a finite number of particles and therefore contain statistical noise.

Finite resolution. Small nodal structures may require very large sample counts before becoming clearly visible.

Single-particle approximation. The project studies only the hydrogen atom. Electron-electron interactions present in multi-electron atoms are not included.

Non-relativistic treatment. The underlying mathematics is based on the non-relativistic Schrödinger equation and therefore neglects relativistic corrections, spin-orbit coupling, and related effects.

10.5 Future Directions

Several natural extensions suggest themselves.

Exact probability-current calculations. Future versions could compute the probability current directly from

$$\mathbf{j} = \frac{\hbar}{2mi} \left(\psi^* \nabla \psi - \psi \nabla \psi^*\right)$$

rather than using the current visualization model.

Phase visualization. Color and transparency could be used to represent wavefunction phase directly, making the complex structure of ψ visible.

Superposition states. The current implementation focuses primarily on individual hydrogenic eigenstates.

Future work could investigate superpositions

$$\Psi = \sum_i c_i \psi_i, \tag{84}$$

allowing interference and time-dependent effects to be explored.

More advanced atomic systems. The framework could be extended beyond hydrogen to investigate multi-electron atoms and approximate quantum-chemical wavefunctions.

11 Conclusion

The Atom project combines analytical quantum mechanics, Monte Carlo sampling, and computer graphics to create an interactive framework for exploring hydrogenic orbitals.
Beginning with exact solutions of the Schrödinger equation, the project constructs numerical representations of orbital probability densities and transforms them into three-dimensional visualizations.
Beyond reproducing familiar orbital shapes, the project provides an opportunity to investigate the distinct roles of the quantum numbers n, l, and m, and to explore the relationship between orbital geometry, angular momentum, phase structure, and probability flow.

Final perspective. The most important lesson of the project is that atomic orbitals are more than geometric objects.

The probability density

$$|\psi|^2$$

describes where an electron is most likely to be observed, but the full wavefunction contains additional structure encoded in its phase.
By visualizing both density and flow, the Atom project attempts to make this distinction visible and to connect the abstract mathematics of the hydrogen atom with an intuitive geometric representation.
In this sense, the project serves not only as a visualization tool, but also as a computational environment for exploring some of the central ideas of quantum mechanics.

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